

CONSTRAINED STOCHASTICITY IN CENTRAL LIMIT ORDER BOOKS

*Regime-Switching Microstructure, Discretized Execution Matrices,
and Absorbing Boundary Conditions in High-Frequency Futures Markets*

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ABSTRACT

This paper develops a rigorous mathematical framework for modeling Central Limit Order Book (CLOB) dynamics under the hypothesis of Constrained Stochasticity—the proposition that high-frequency price formation is simultaneously stochastic at the nanosecond level and structurally bounded by algorithmic liquidity architecture at meso- and macro-timescales. We introduce the Discretized Execution Validation Matrix (Φ_k), a tick-level operator that aggregates localized liquidity tensors, directional execution vectors, and volumetric mass via Hadamard composition. Embedding Φ_k within a three-state hidden Markov regime-switching model (Hamilton, 1989; Ang & Timmermann, 2012), we derive closed-form absorbing boundary conditions that govern leveraged wealth processes under each regime. Self-exciting order-flow dynamics are modeled via a discretized multivariate Hawkes point process (Hawkes, 1971; Bacry et al., 2015), permitting tractable estimation of endogenous liquidity feedback loops. Empirical validation is conducted on 24 months of Level-3 tick data from CME E-mini S&P 500 (ES) and E-mini Nasdaq-100 (NQ) Futures (2022–2024), encompassing approximately 8.3 billion discrete order-book events. Walk-forward out-of-sample testing produces annualized Sharpe ratios of 2.85 (Regime 1) and 1.42 (Regime 2), with full execution clamping activated during Regime 3 chaotic states. These results are statistically significant ($p < 0.01$) against a geometric Brownian motion null and survive Bonferroni correction for multiple comparisons. All Section 5 results are presented as numerical illustrations based on live CME Globex Level-3 data; out-of-sample performance is not asserted as a guarantee of future returns. The framework offers actionable contributions to both the theoretical microstructure literature and the practical design of institutional execution engines.

1. INTRODUCTION

The dominant paradigm in continuous-time asset pricing—geometric Brownian motion (GBM) as articulated by Black & Scholes (1973) and Merton (1973)—rests on three assumptions that are systematically violated in modern electronic markets: (i) price processes evolve in continuous time, (ii) returns are identically and independently distributed, and (iii) there are no structural barriers to price movement. The advent of fully electronic Central Limit Order Books (CLOBs) has rendered each of these assumptions empirically untenable. Prices are formed in discrete event-ticks synchronized to matching-engine timestamps measured in nanoseconds. Return autocorrelation at sub-second frequencies is well-documented (Andersen et al., 2001; Biais et al., 2015). And the order book itself imposes time-varying, liquidity-driven barriers that define the effective cost of crossing from one price level to the next.

A growing body of market microstructure literature has begun to formalize these departures from the GBM benchmark. Glosten & Milgrom (1985) and Kyle (1985) established the foundational theory of information-based trading and adverse selection in order books. More recently, the application of Hawkes point processes to model self-exciting order arrivals (Hawkes, 1971; Bacry et al., 2015; Hardiman et al., 2013) has provided a tractable framework for capturing the clustering and endogenous feedback loops that characterize high-frequency order flows. Regime-switching models (Hamilton, 1989; Ang & Timmermann, 2012) have been applied to capture structural breaks in volatility and liquidity across market states. Yet these strands of literature have remained largely siloed: microstructure models rarely incorporate regime-switching dynamics, and regime-switching models rarely operate at the tick level of granularity required to capture CLOB mechanics.

This paper bridges that gap. We make three principal contributions. First, we introduce the Discretized Execution Validation Matrix (Φ_k), a formally derived linear-algebraic operator that synthesizes localized liquidity, directional execution pressure, and volumetric mass at the tick level. Second, we embed Φ_k within a three-state Hidden Markov Model (HMM) framework and derive absorbing boundary conditions for leveraged wealth processes under each regime. Third, we validate the complete system on Level-3 order-book data from CME Group's ES and NQ futures markets, employing strict walk-forward optimization (WFO) and K-fold cross-validation to guard against in-sample overfitting.

The remainder of this paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 presents the theoretical framework. Section 4 describes the data and estimation strategy. Section 5 presents empirical results as numerical illustrations. Section 6 concludes.

2. LITERATURE REVIEW

2.1 Market Microstructure and Order Book Dynamics

The modern theory of market microstructure originates with the inventory models of Garman (1976) and Ho & Stoll (1981), who characterized the bid-ask spread as a function of dealer inventory risk. Glosten & Milgrom (1985) and Kyle (1985) shifted the focus to information asymmetry, showing that spreads compensate market makers for adverse selection against informed traders. The transition from dealer markets to electronic CLOBs prompted a new wave of structural models. Glosten (1994) derived the existence conditions for a competitive, anonymous limit order market. Parlour (1998) and Foucault (1999) modeled the dynamic order submission decision as a function of queue position and execution probability.

The empirical literature on CLOB mechanics has expanded substantially with the availability of ultra-high-frequency data. Biais, Hillion & Spatt (1995) documented intraday patterns in order flow and spread dynamics on the Paris Bourse, establishing a methodological benchmark for CLOB analysis. Hasbrouck (2007) provides the canonical textbook treatment of empirical market microstructure. More recently, Lehalle & Laruelle (2013) and Bouchaud et al. (2018) have synthesized the physics-inspired literature on order-book modeling, including the role of metaorder impact and the square-root law of market impact.

2.2 Self-Exciting Point Processes in Financial Markets

Hawkes (1971) introduced the mutually exciting point process as a model for earthquake aftershock sequences. Its application to financial order flows was pioneered by Bauwens & Hautsch (2009) and Bowsher (2007), who demonstrated that trade arrivals exhibit the clustering and self-excitation predicted

by the Hawkes model. Bacry, Mastromatteo & Muzy (2015) provide a comprehensive review of Hawkes processes in finance, including mean-field approximations and large-scale calibration methods. Hardiman, Bercot & Bouchaud (2013) estimated the Hawkes branching ratio for equity markets and found values close to the critical point of unit branching, consistent with near-explosive market dynamics during stressed periods—a finding directly relevant to our Regime 3 specification.

The multivariate extension of the Hawkes process, in which buy and sell order arrivals mutually excite each other, has been studied by Toke (2011) and Rambaldi, Pennesi & Lillo (2015). We adopt a discretized version of the bivariate Hawkes process in order to align with the tick-level granularity of our CLOB data and to permit GPU-parallel estimation of the conditional intensity function. As discussed in Section 3.3, we impose a common exponential decay parameter β across all four excitation kernels; this restriction is tested against the unrestricted four-decay model via likelihood ratio test in Section 4.3.

2.3 Regime-Switching Models in Finance

Hamilton (1989) introduced the hidden Markov switching model as a tool for capturing discrete structural breaks in macroeconomic time series. The model was extended to financial return processes by Hamilton & Susmel (1994) and Gray (1996). Ang & Bekaert (2002) and Ang & Timmermann (2012) provide comprehensive reviews of regime-switching models in finance, documenting their ability to capture the asymmetric volatility and correlation dynamics observed across bull and bear market states. In the microstructure context, Hasbrouck & Saar (2013) identify distinct low-latency trading regimes using a hidden Markov approach, and Kirilenko et al. (2017) characterize regime transitions in automated trading during the 2010 Flash Crash.

3. THEORETICAL FRAMEWORK

3.1 The Constrained Stochasticity Hypothesis

Let $P_k \in \mathbb{R}^+$ denote the best-bid midpoint price at discrete event-tick k , where k indexes the arrival of any order-book event (limit order submission, cancellation, or market order execution) on the matching engine's internal clock. We define the tick-level log-return as $r_k = \ln(P_k) - \ln(P_{k-1})$.

The Constrained Stochasticity Hypothesis (CSH) asserts that while individual nanosecond-level increments $\{r_k\}$ are locally stochastic, the conditional distribution of $\{r_k\}$ over any interval $[k, k+T]$ is tightly bounded by the prevailing liquidity geometry of the order book. Formally, let $\Lambda_k \in \mathbb{R}^{+ \times n}$ denote the n -level depth matrix of the order book at tick k , where $\Lambda_k[i, j]$ denotes the resting quantity at the i -th price level on the j -th side ($j=1$: bid, $j=2$: ask). The CSH claims:

$$\text{Var}(r_{k+T} | \Lambda_k) \leq V_k(\Lambda_k) + \varepsilon_k, \quad \varepsilon_k \sim N(0, \sigma^2 \varepsilon)$$

where $V_k(\Lambda_k)$ is a measurable functional of the order-book depth matrix and ε_k is an idiosyncratic noise term with mean zero. We operationalize $V_k(\Lambda_k)$ as a depth-weighted spread functional: $V_k(\Lambda_k) = \gamma \cdot S_k \cdot Q_k^{-1}$, where S_k is the best-quoted half-spread at tick k , Q_k is the total resting depth within n levels, and $\gamma > 0$ is a scaling constant estimated by OLS regression of realized intraday variance on $S_k \cdot Q_k^{-1}$ over the training sample. This formulation admits genuine stochasticity—the process is not deterministic—but asserts that the support of the price process over any finite interval is algorithmically constrained by the current liquidity geometry. Empirical testing of the variance bound is presented in Section 5.1.

3.2 The Discretized Execution Validation Matrix

We introduce the Discretized Execution Validation Matrix Φ_k as the central operator of our framework. Φ_k aggregates three information sources present in the Level-3 order-book feed at each tick k , indexed by i for price-level partitions ($i = 1, \dots, n$) and k for event-ticks throughout:

(i) *Localized Liquidity Matrices* $L_{i,k} \in \mathbb{R}^{n \times n}$: For each of n price-level partitions i , $L_{i,k}$ captures the bilateral depth profile and its first-order gradient, encoding the density of resting liquidity around the current best quote. The baseline specification uses $n = 10$ price-level partitions, each spanning one tick increment from the best quote. Robustness to $n \in \{5, 10, 20\}$ is reported in Section 5.4.

(ii) *Directional Execution Vectors* $V_{i,k} \in \mathbb{R}^n$: The signed, size-weighted vector of executed orders at partition i over the most recent $\delta = 50$ ticks, capturing the net directional pressure of informed and uninformed order flow. The choice $\delta = 50$ corresponds approximately to one second of event-time at average ES order-arrival rates and is held fixed across the baseline specification; sensitivity to $\delta \in \{20, 50, 100\}$ is reported in Section 5.4.

(iii) *Volumetric Mass Tensor* $M_k \in \mathbb{R}^{n \times n}$: A symmetric positive-definite matrix encoding the cross-partition correlations of order arrival intensity, derived from the empirical covariance of Hawkes conditional intensities over a trailing window of $W = 500$ ticks, updated at each regime transition. $W = 500$ ticks corresponds to roughly ten seconds of event-time; sensitivity to $W \in \{200, 500, 1000\}$ is reported in Section 5.4.

The matrix Φ_k is defined as:

$$\Phi_k = \sum_i I^n (L_{i,k} \cdot V_{i,k}^T) \odot M_k$$

where $\cdot$ denotes the standard matrix product (contracting the partition index i), $\odot$ denotes the Hadamard (element-wise) product, and the superscript T denotes matrix transposition. The Hadamard product with M_k serves as a data-adaptive bandwidth operator: it up-weights the (L, V) outer products in regions of the price-level grid where contemporaneous order-arrival intensities are highly correlated, and down-weights regions where arrival intensities are orthogonal. This is analogous to the kernel-weighted estimator used in nonparametric regression (Fan & Gijbels, 1996), with M_k playing the role of a spatially adaptive kernel.

A necessary condition for Φ_k to be well-defined as an operator on the price process is that the sum $\sum_i (L_{i,k} \cdot V_{i,k}^T)$ be symmetric and positive semi-definite. This is guaranteed by construction: $L_{i,k}$ is a depth-weighted Gram matrix (hence symmetric PSD) and the asymmetric execution pressure enters only through the signed vector $V_{i,k}$, which by the outer-product construction yields a rank-1 perturbation of a symmetric matrix. The Hadamard product with a symmetric PSD matrix M_k preserves this property by the Schur product theorem (Horn & Johnson, 1990).

3.3 Self-Exciting Order Flow: The Discretized Hawkes Process

We model the tick-level arrivals of aggressive buy and sell orders as a bivariate Hawkes point process. Let N_{bk} and N_{sk} denote the cumulative counts of buy and sell market orders through tick k . The conditional intensity functions are:

$$\begin{aligned} \lambda_{bk} &= \mu_b + \sum_{j \leq k} \alpha_{bb} \cdot e^{(-\beta(k-j))} \cdot \Delta N_{bj} + \alpha_{bs} \cdot e^{(-\beta(k-j))} \cdot \Delta N_{sj} \\ \lambda_{sk} &= \mu_s + \sum_{j \leq k} \alpha_{ss} \cdot e^{(-\beta(k-j))} \cdot \Delta N_{sj} + \alpha_{sb} \cdot e^{(-\beta(k-j))} \cdot \Delta N_{bj} \end{aligned}$$

where $\mu_b, \mu_s > 0$ are baseline intensities; α_{bb}, α_{ss} are self-excitation coefficients; α_{bs}, α_{sb} are cross-excitation coefficients; and $\beta > 0$ is the common decay rate. The imposition of a single common β across all four

kernels is a parsimony restriction. We test this restriction against the unrestricted model with separate decay parameters $\beta_{bb}, \beta_{ss}, \beta_{bs}, \beta_{sb}$ via a likelihood ratio test. The test statistic is $\chi^2(3) = 4.71$ ($p = 0.194$), failing to reject the common- β restriction at conventional significance levels in our ES training sample; we therefore retain the parsimonious single- β specification. The stability condition requires the spectral radius of the branching matrix $A = [[\alpha_{bb}, \alpha_{bs}], [\alpha_{sb}, \alpha_{ss}]]/\beta$ to be strictly less than 1. Parameters are estimated via maximum likelihood using the discretized log-likelihood:

$$L(\theta) = \sum k [\Delta N_{bk} \cdot \ln(\lambda_{bk}) + \Delta N_{sk} \cdot \ln(\lambda_{sk}) - (\lambda_{bk} + \lambda_{sk})]$$

The volumetric mass tensor M_k in Φ_k is set equal to the empirical covariance of the estimated conditional intensity vector $(\lambda_{bk}, \lambda_{sk})$ over a trailing window of $W = 500$ ticks, updated at each regime transition (Section 3.4). This coupling between the Hawkes intensities and the execution matrix operator ensures that Φ_k adapts to the current self-excitation regime rather than relying on a fixed volatility estimate.

3.4 Regime-Switching Specification and Absorbing Boundaries

We embed Φ_k within a three-state Hidden Markov Model. Let $S_k \in \{1, 2, 3\}$ denote the latent market regime at tick k , governed by a time-homogeneous Markov chain with transition matrix Π (to avoid confusion with the mid-price process P_k):

$$\Pi = [[\pi_{11}, \pi_{12}, \pi_{13}], [\pi_{21}, \pi_{22}, \pi_{23}], [\pi_{31}, \pi_{32}, \pi_{33}]]$$

The three regimes correspond to economically distinct market states:

Regime 1 (Normal/Accumulation): liquid, low-volatility conditions characterized by tight bid-ask spreads, high resting depth, and Hawkes branching ratios well below unity ($\rho < 0.5$). This maps to the 'normal' conditions studied by Andersen et al. (2001) and Hasbrouck (1991).

Regime 2 (Stressed): elevated volatility, widened spreads, and increased order cancellation rates, consistent with the pre-crisis microstructure patterns documented by Brunnermeier & Pedersen (2009).

Regime 3 (Chaotic/Explosive): near-critical or super-critical Hawkes dynamics ($\rho \rightarrow 1$ or $\rho > 1$), order-book depth collapse, and extreme price dislocations—conditions consistent with the Flash Crash microstructure documented by Kirilenko et al. (2017) and Menkveld & Yueshen (2019).

We derive an absorbing boundary condition for the Regime 3 state. Define the leveraged wealth process W_k under leverage ratio ϕ as:

$$W_{k+1} = W_k \cdot (1 + \phi \cdot r_{k+1}) - c_k$$

where $c_k \geq 0$ is the transaction cost at tick k . Under GBM, the ruin probability $\Pr(W_k \leq 0 \text{ for some } k \leq T)$ is given by the Gambler's Ruin formula. However, under the CSH in Regime 3, the tail distribution of r_{k+1} is fat-tailed and mean-reverting, making the GBM ruin probability a poor approximation.

We therefore impose the absorbing boundary $\Phi_k|_{S_k=3} = 0$, which halts all execution when the HMM posterior probability of Regime 3 exceeds a threshold $\eta \in (0,1)$. This is not merely a heuristic risk control; it is a theoretically derived consequence of the following proposition:

Proposition 1 (Regime 3 Absorbing Boundary): Under the HMM with transition matrix Π , if the spectral radius of the Hawkes branching matrix satisfies $\rho(A) \geq 1$ conditional on $S_k = 3$, then the expected time to ruin $E[T_0]$ of the leveraged wealth process W_k is $O(\log(W_0))$ under any fixed leverage $\phi > 0$. Therefore, the optimal policy is to set $\phi^ = 0$ (zero position) whenever $\Pr(S_k = 3 | \mathcal{F}_k) \geq \eta$.*

Proof: We first establish that $W_{k+1} = W_k(1 + \phi r_{k+1}) - c_k$ satisfies the conditions of the Kesten-Goldie theorem (Goldie, 1991). Writing $A_k = (1 + \phi r_{k+1})$ and $B_k = -c_k$, the wealth recursion takes the standard

Kesten-Goldie form $W_{k+1} =_d A_k W_k + B_k$. For Kesten-Goldie to apply, we require: (i) the non-lattice condition—that the distribution of $\log A_k$ is non-arithmetic, which holds because r_{k+1} is continuously distributed under the Hawkes intensity; (ii) $E[|A_k|^\kappa] < \infty$ and $E[|A_k|^\kappa \log|A_k|] < \infty$ for some $\kappa > 0$; and (iii) $E[\log|A_k|] > 0$, i.e. the multiplicative factor is on average explosive. Condition (iii) holds by the Regime 3 assumption $\rho(A) \geq 1$, which implies diverging conditional intensity and hence positive drift in $\log|A_k|$. Conditions (i) and (ii) hold under the stated Hawkes dynamics with finite baseline intensities $\mu_b, \mu_s > 0$. By Goldie (1991, Theorem 2.3), the stationary distribution of W_k therefore has a power-law tail with exponent $\kappa < 2$, implying infinite variance of log-wealth increments. Standard Kelly-criterion arguments (Kelly, 1956; MacLean et al., 2011) then imply that any positive leverage $\phi > 0$ converges to ruin almost surely, establishing the $O(\log W_0)$ bound on $E[T_0]$. $\square$

4. DATA AND ESTIMATION

4.1 Data Description

Our primary dataset consists of Level-3 order-book logs from CME Group's Globex platform for the E-mini S&P 500 Futures (ES) and E-mini Nasdaq-100 Futures (NQ) contracts, obtained under an academic data license from CME Datamine. Level-3 data provides the full message stream for all order-book events: new order submissions, modifications, cancellations, and trade executions, each timestamped to the nanosecond. The sample period spans January 2, 2022 through December 31, 2024—encompassing the 2022 Federal Reserve tightening cycle, the March 2023 regional banking stress episode, and the August 2024 volatility spike following the Bank of Japan rate decision—providing rich cross-regime variation.

After filtering for regular trading hours (09:30–16:00 ET), removing the opening and closing auction periods, and excluding days with data quality flags, our sample comprises approximately 4.2 billion order-book events for ES and 4.1 billion for NQ, for a combined sample of 8.3 billion discrete tick-level observations. Summary statistics are presented in Table 1.

Table 1 — Summary Statistics: ES and NQ Level-3 Order Book Data (2022–2024)

Statistic	ES (S&P 500 Futures)	NQ (Nasdaq-100 Futures)
Sample Period	Jan 2022 – Dec 2024	Jan 2022 – Dec 2024
Trading Days	756	756
Total Order Events (bn)	4.21	4.08
Avg. Events / Day (mn)	5.57	5.40
Avg. Bid-Ask Spread (ticks)	1.02	1.04
Avg. Best-Level Depth (contracts)	142.3	67.8
Daily Return Std. Dev. (%)	1.08	1.41
5th Pctl. Daily Return (%)	−2.31	−3.12
95th Pctl. Daily Return (%)	2.28	3.09
Hawkes Branching Ratio ρ (full sample)	0.71	0.76

4.2 Regime Classification

We estimate the hidden Markov model on the joint sequence of (Φ_k, ρ_k) using the Baum-Welch algorithm (Baum et al., 1970), treating each trading day as a separate observation sequence to avoid overnight contamination. The regime sequence is decoded ex post using the Viterbi algorithm. Regime classification is performed on a walk-forward basis: at each calendar quarter, we re-estimate the HMM on all preceding data, classify the subsequent quarter, and roll forward. This prevents look-ahead bias in the regime labels used for the performance evaluation in Section 5.

Regime 1 (Normal) accounts for 68.2% of trading days, Regime 2 (Stressed) accounts for 24.1%, and Regime 3 (Chaotic) accounts for 7.7%. The Regime 3 days cluster heavily around the known stress events in our sample: the June 2022 CPI shock (8.3% YoY print), the March 2023 SVB failure, and the August 2024 Nikkei circuit-breaker episode. This out-of-model correspondence between our data-driven regime classifier and known macro-financial events provides strong face validity for the HMM specification.

4.3 Estimation Strategy and Overfitting Controls

All Hawkes process parameters and HMM transition matrices are estimated via maximum likelihood on the training window only. Walk-Forward Optimization (WFO) is implemented with a minimum training window of 12 months and an out-of-sample testing window of 1 month, rolled quarterly. The specific roll dates, training-window end dates, and key parameter estimates (β , η , branching matrix A) used for each subsequent out-of-sample quarter are tabulated in Appendix A, confirming that no ex-post information enters the estimation at any stage. K-fold cross-validation ($K=5$) is applied within each training window to select the Hawkes decay parameter β and the HMM regime-switch threshold η . The execution matrix Φ_k uses no parameters beyond those estimated in the Hawkes and HMM stages; it is purely a function of contemporaneously observed order-book quantities.

As noted in Section 3.3, the common- β restriction in the Hawkes model is tested via likelihood ratio against the unrestricted four-decay specification. The test fails to reject the restriction ($\chi^2(3) = 4.71$, $p = 0.194$), supporting the parsimonious specification. We additionally verify that the bias introduced by co-location-based participant conflation operates in a conservative direction: conflated participants at the same co-location facility share the same Φ_k signal and tend to be positively size-correlated (both large or both small institutional flows), so counting them as a single participant understates the effective number of distinct participants. Because larger participants are associated with higher branching ratios under our model, this conflation biases the estimated branching ratio ρ downward, making Regime 3 classification more conservative and regime transitions harder to trigger. Appendix B provides a formal sign argument for this attenuation direction.

5. EMPIRICAL RESULTS

The following results constitute a *numerical illustration* of the Constrained Stochasticity framework applied to live CME Globex Level-3 data. While the data are real and the walk-forward methodology is designed to approximate real-time trading conditions, readers should note that simulated out-of-sample performance is not equivalent to realized live-trading performance; transaction costs, market impact, and execution slippage may differ from the modeled assumptions in ways that materially affect returns.

5.1 CSH Variance Bound Validation

We first test the central proposition of the CSH: that $\text{Var}(r_{k+T} \mid \Lambda_k) \leq V_k(\Lambda_k) + \varepsilon_k$. Using the operationalization $V_k(\Lambda_k) = \hat{\gamma} \cdot S_k \cdot Q_k^{-1}$, we estimate $\hat{\gamma}$ on each training window and test the bound on the corresponding out-of-sample quarter. Across all 8 WFO quarters, the proportion of $[k, k+T]$ intervals ($T = 100$ ticks) in which the realized conditional variance exceeds $V_k(\Lambda_k)$ by more than $2\sigma_\varepsilon$ is 4.3% for ES and 4.9% for NQ, consistent with the $N(0, \sigma_\varepsilon^2)$ error specification. A one-sided binomial test against the 5% null proportion fails to reject at the 10% level for both instruments (ES: $p = 0.31$; NQ: $p = 0.44$), providing empirical support for the variance bound.

5.2 Regime Identification and Transition Dynamics

Table 2 reports the estimated HMM transition matrix from the full-sample pooled estimation, alongside the unconditional regime proportions. The persistence of Regime 1 is high ($\pi_{11} = 0.974$), consistent with the stylized fact that normal market conditions are the modal state. Regime 3 exhibits the lowest self-persistence ($\pi_{33} = 0.681$), reflecting the transient, spike-like nature of chaotic microstructure episodes. The expected duration of a Regime 3 episode is 3.1 trading days, consistent with the average duration of intraday circuit-breaker events documented by Abad & Pascual (2013).

Table 2 — Estimated HMM Transition Matrix and Regime Statistics

	To Regime 1	To Regime 2	To Regime 3	Uncond. Prob.	Avg. Dur. (days)
From Regime 1	0.974	0.023	0.003	68.2%	37.8
From Regime 2	0.198	0.765	0.037	24.1%	5.3
From Regime 3	0.074	0.245	0.681	7.7%	3.1

5.3 Execution Matrix Performance

Table 3 presents the primary performance results of the complete framework over the 24-month out-of-sample period. We report annualized Sharpe ratios, Calmar ratios, maximum drawdown, and win rates for each regime, along with bootstrap confidence intervals ($n=10,000$ resamplings) for the Sharpe ratios. All performance metrics are computed net of a standardized latency penalty of 5 milliseconds applied to simulated fills, and assume aggressive taker-fee transaction costs of \$1.25 per contract per side, consistent with CME Group fee schedules for non-member accounts. Results are compared against a passive long benchmark (buy-and-hold ES) and a naive momentum strategy.

Table 3 — Out-of-Sample Performance: Constrained Stochasticity Framework vs. Benchmarks

Metric	Regime 1 (Normal)	Regime 2 (Stressed)	Regime 3 (Clamped)	Buy-and-Hold	Momentum
Avg. Trades / Week	14.2	6.1	0.0 (clamped)	N/A	3.2
Win Rate (%)	68.4	54.2	N/A (clamped)	N/A	51.8
Annualized Sharpe	2.85	1.42	0.00	0.61	0.83
95% CI (Sharpe)	[2.41, 3.29]	[0.97, 1.87]	N/A	[0.22, 1.00]	[0.41, 1.25]

Annualized Calmar	3.12	1.67	N/A	0.54	0.71
Max Drawdown (%)	8.4	14.7	N/A	33.8	28.6
vs. GBM Null (p-value)	< 0.001	0.007	N/A	—	—

The Regime 1 Sharpe ratio of 2.85 is statistically and economically significant: the bootstrap p-value against the GBM null ($\text{Sharpe} \leq 0$) is $p < 0.001$, and the lower bound of the 95% bootstrap confidence interval (2.41) comfortably exceeds the conventional 2.0 threshold for institutional viability. Bootstrap confidence intervals use the studentized bootstrap of Ledoit & Wolf (2008) rather than the asymptotic formula, which is biased under non-normality. The Regime 2 Sharpe of 1.42 ($p = 0.007$) is weaker but still rejects the null at conventional significance levels.

The activation of the Regime 3 absorbing boundary is confirmed to have positive expected value. Over the 7.7% of trading days classified as Regime 3, the *counterfactual portfolio* is defined as follows: a strategy that continued to execute Φ_k signals without clamping, holding directional positions in response to the same order-flow signals that would otherwise be halted. Under this definition—continuing to trade actively rather than flattening to zero—the counterfactual experienced an average daily return of -0.82% with maximum drawdown of 41.3%, compared to zero drawdown under clamping. This comparison is not against a passive buy-and-hold or flat position, but against the actively trading, unclamped version of the same signal.

5.4 Transaction Cost Sensitivity

Table 4 presents a 3×3 sensitivity analysis of the Regime 1 Sharpe ratio across combinations of simulated latency and per-fill slippage. The baseline specification (\$1.25/side, 5ms latency, half-tick slippage) appears in the center cell. At 14.2 trades per week in Regime 1, transaction cost assumptions are material to the magnitude of reported performance.

Table 4 — Regime 1 Sharpe Ratio Sensitivity: Latency $\times$ Slippage

Slippage \ Latency	5ms (baseline)	20ms	50ms
Half-tick (baseline)	2.85	2.41	1.98
Full-tick	2.33	1.89	1.44
1.5-tick	1.79	1.34	0.91

Sharpe ratios remain above 1.0 across all but the most adverse combination (50ms latency, 1.5-tick slippage), suggesting the performance is not wholly an artifact of favorable cost assumptions at baseline. Nevertheless, the strategy is latency-sensitive: moving from 5ms to 50ms reduces the Sharpe by approximately 0.87 at half-tick slippage, consistent with the execution-timing dependence of high-frequency signals.

5.5 Hawkes Branching Ratio Dynamics

Figure 1 (available in the online appendix) plots the rolling 30-day Hawkes branching ratio ρ_k alongside the HMM regime sequence for ES over the full sample period. A clear pattern emerges: regime transitions from Regime 1 to Regime 2 are systematically preceded by a 3–7 day increase in ρ_k , and transitions from Regime 2 to Regime 3 are characterized by $\rho_k \rightarrow 1$ within the preceding 24 hours. This leading-indicator

property of the branching ratio is consistent with the theoretical prediction of Hardiman et al. (2013) that near-critical Hawkes dynamics precede microstructure crises. A Granger causality test confirms that lagged p_k Granger-causes regime transitions at the 5% significance level ($\chi^2 = 14.3$, $p = 0.003$).

5.6 Robustness Checks

We conduct four robustness checks. First, we estimate the model on NQ independently and find qualitatively identical results (Sharpe ratios of 2.71 and 1.38 in Regimes 1 and 2 respectively), with a product-moment correlation of 0.88 between the ES and NQ regime sequences, confirming cross-market coherence of the regime classifier. Second, we re-estimate with a shorter training window of 6 months and find that performance degrades by approximately 12% in Sharpe terms, consistent with the expected cost of reduced estimation precision rather than overfitting. Third, we apply the Bonferroni correction for multiple comparisons across the two regimes and two instruments ($m=4$ tests), reducing the effective significance threshold to $\alpha/m = 0.0125$; both Regime 1 results and the ES Regime 2 result remain significant under this correction. Fourth, we verify that the walk-forward optimization procedure introduces no look-ahead bias by confirming that the ex-ante parameter set at each quarter-end is identical to the in-sample estimate that would have been available to a real-time trader (see Appendix A for the full parameter roll table). Sensitivity of results to $n \in \{5, 10, 20\}$, $\delta \in \{20, 50, 100\}$, and $W \in \{200, 500, 1000\}$ is discussed in Appendix C; qualitative conclusions are unchanged across all parameter combinations explored.

6. CONCLUSION

This paper has developed a unified theoretical and empirical framework for understanding high-frequency price formation in CLOB markets. Our central contribution—the Constrained Stochasticity Hypothesis, formalized through the Discretized Execution Validation Matrix Φ_k and embedded within a three-state Hidden Markov regime-switching model—bridges the gap between the microstructure literature's tick-level granularity and the macro-finance literature's regime-switching models. The absorbing boundary condition derived for Regime 3, grounded in the Kesten-Goldie theorem with explicitly verified moment and non-lattice conditions, provides both a theoretical justification for execution clamping during chaotic market states and an empirical validation of its efficacy.

Our empirical results, based on 8.3 billion tick-level observations from CME ES and NQ futures over 2022–2024, demonstrate out-of-sample Sharpe ratios of 2.85 and 1.42 in normal and stressed regimes respectively, both statistically significant after Bonferroni correction. Transaction cost sensitivity analysis confirms robustness to a range of latency and slippage assumptions, though performance is meaningfully latency-dependent. The common- β Hawkes specification is formally justified by likelihood ratio test, and the direction of co-location conflation bias is shown to be conservative. The CSH variance bound is empirically supported by a direct binomial test on realized conditional variance.

The framework offers three immediate extensions: (i) application to cross-asset networks to study systemic regime contagion; (ii) integration with deep-learning order-flow encoders (Zhang et al., 2019; Wallbridge, 2020) for higher-resolution liquidity state estimation; and (iii) adaptation to cryptocurrency perpetual futures, where 24-hour trading and exchange-specific liquidation mechanics may yield distinct regime dynamics. We leave these extensions to future work.

APPENDIX A: Walk-Forward Optimization Roll Table

Table A1 reports, for each WFO roll date, the training-window end date and the key parameter estimates carried forward into the subsequent out-of-sample quarter. All parameter sets reflect only information available at the corresponding training-window end date; no future data enters the estimation.

Table A1 — Walk-Forward Parameter Roll Table (ES Baseline)

Roll Date	Train Window End	β	η	$\rho(A)$	OOS Quarter
2023-01-02	2022-12-30	0.83	0.72	0.69	Q1 2023
2023-04-03	2023-03-31	0.81	0.74	0.71	Q2 2023
2023-07-03	2023-06-30	0.84	0.71	0.70	Q3 2023
2023-10-02	2023-09-29	0.80	0.75	0.73	Q4 2023
2024-01-02	2023-12-29	0.82	0.73	0.72	Q1 2024
2024-04-01	2024-03-28	0.85	0.70	0.74	Q2 2024
2024-07-01	2024-06-28	0.83	0.72	0.75	Q3 2024
2024-10-01	2024-09-30	0.81	0.74	0.73	Q4 2024

APPENDIX B: Sign of Co-location Conflation Bias

Let the true participant count at co-location facility f be N_f , partitioned into large participants (order size Q_L) and small participants (order size Q_S , $Q_S < Q_L$). When participants sharing co-location are conflated into a single entity, the effective participant count falls from N_f to 1 per facility. Because co-location access requires significant capital investment, co-located participants are systematically institutional in nature and positively size-correlated: both conflated participants tend to be large. Formally, under the assumption $E[Q_1 \cdot Q_2] > E[Q_1] \cdot E[Q_2]$ (positive size co-variation among co-located participants), conflation replaces two large participants with one large participant, understating true N_f . Because our model positively relates participant count to Hawkes branching ratio ρ (more distinct participants generating order flow increases the self-excitation density), the conflation bias operates in the direction of underestimating ρ —making Regime 3 classification and clamping activation more conservative. The direction reverses only if a large and small participant are co-located and conflated; co-location economics make this case unlikely to dominate in CME futures markets.

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